

Protected vacua, shared mediators and the alignment boundary

Singlet vacuum dynamics, canonical quark matching and explicit limits on flavor alignment

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The selected model admits controlled vacua near ± 66 degrees. Its initialized equations receive perturbative supersymmetric protection; its energy selection, physical coefficient readout and quark couplings remain separately specified interactions.

Exact rational results and numerical scenario results are distinguished throughout. The worked construction is conditional and does not claim an independently derived theory of flavor.

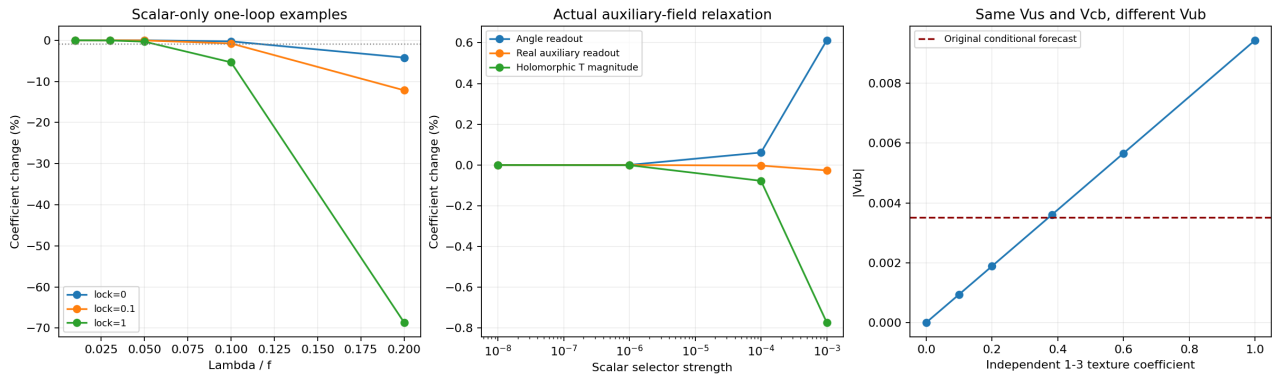
What this push accomplishes

The subsequent shared-mediator investigation is documented in B5_FLAVOR_MEDIATOR.md and included as the second part of the reader monograph. It derives exact canonical tree matching, establishes a scoped low-degree alignment obstruction, and tests an adjoint alternative. It transmits an initialized golden frame but does not derive that frame from symmetry.

The candidate now has an explicit renormalizable supersymmetric singlet sector, a holomorphic golden-coefficient extension, an actual auxiliary-field relaxation calculation, and a one-loop calculation including every scalar and fermion in that sector. The 34-field extension has 68 real scalar modes and 34 Weyl fermions. In the declared numerical examples, a small scalar energy selector prefers the branches near plus and minus 66 degrees, and the one-loop corrections leave that preference intact among all sixteen continued branches. The construction produces a quantitatively controlled realization of the nominated vacuum. It does not derive the initial superpotential coefficients or the energy-selector pattern from an independently specified symmetry, and it does not derive the nominated exact CKM relation from quark interactions.

The strongest negative result is exact: the scalar-only version of the potential is not closed under its own one-loop divergent counterterm. Polynomial division yields a nonzero torque remainder, and a Bezout identity shows that the remainder vanishes at none of the primitive real order-60 roots. Thus there is a specific radiative protection problem, even before quarks or gauge fields are introduced. The strongest constructive result is an initialized holomorphic constraint circuit whose supersymmetric zeros are perturbatively protected. Soft selection then produces small calculable departures instead of an exact unchanged angle.

The quark calculation exposes another independent issue. The declared flavor charges allow the required hierarchy of operators, but their coefficients are independent. Exact diagonalization of a selected triangular texture, even with its 1-3 coefficient initialized to the golden value, gives a different observable mixing strength. This is a constructive counterexample to assuming that the right spurion powers automatically imply the exact CKM formula.



The original exact carrier and energy winner

$$V'_{\text{lock}}(x) = P(x) [2\lambda P'(x) + \eta Q(x)]$$

Define $x = 2 \cos(\theta)$, $P(x) = x^8 - 7x^6 + 14x^4 - 8x^2 + 1$ and $Q(x) = -1 + 3x - x^3$. The previously constructed potential obeys $V'(x) = P(x)Q(x)$. Its six nonconstant angular terms are $\cos(2$

$\theta) + (2/3)\cos(3\theta) + (2/5)\cos(5\theta) + (1/4)\cos(8\theta) - (2/9)\cos(9\theta) - (1/6)\cos(12\theta)$. V differs from that angular expression by the constant $11/12$. Exact rational Sturm isolation and energy enclosures selected $x=2\cos(66^\circ)$ on $[-2,2]$, with more than 0.060 separation from the next competing stationary energy in the polynomial normalization.

The new family $V_{\text{lock}}(x) = \lambda P(x)^2 + \eta V(x)$, $\lambda \geq 0$ and $\eta > 0$, obeys $V'_{\text{lock}}(x) = P(x)[2\lambda P'(x) + \eta Q(x)]$. Its global winner is unchanged: the added square is nonnegative, vanishes at the old winner, and the original potential already has a unique global winner in x . This statement covers the full declared parameter orthant without discretizing λ or η . The two full-circle CP partners remain degenerate. This is a mathematical family, not an assertion that the interaction generating λ or η has been identified.

The rational stationary-family constraint is stronger than stationarity under arbitrary real couplings at just one chosen angle. Rational coefficients force Galois-conjugate roots to share the polynomial factor. An isolated real linear torque condition at a prescribed angle is not equivalent to the eight rational relations found for the twelve-harmonic carrier. The field model must explain which of those structures it actually enforces.

A precise scalar-loop obstruction

$$h(x) = (4 - x^2)V''(x) - xV'(x), \quad \gcd(P, (h^2)') = 1$$

Give the angle the kinetic term $f^2 (\partial \theta)^2$ and the physical potential $\Lambda^4 V(2\cos(\theta))$. With $D(x) = 4 - x^2$, its dimensionless angular curvature is $h(x) = D(x)V''(x) - xV'(x)$. The physical field-dependent squared mass is $\Lambda^4 h(x)/f^2$. The one-loop scalar divergence contains a coefficient proportional to this mass to the fourth power. Consequently the question of preserving the stationary family reduces to whether P divides the x derivative of h^2 .

For the original potential, the exact remainder is

$R(x) = 5040 - 6880x - 31360x^2 + 43600x^3 + 24160x^4 - 30080x^5 - 4400x^6 + 4800x^7$. Its $\gcd$ with P is one. The stored rational Bezout multipliers satisfy $a(x)P(x) + b(x)R(x) = 1$. Since $\sin(\theta)$ is nonzero at every primitive order-60 angle, a nonzero x torque also means a nonzero angular torque. This establishes that the original family is not closed under this particular divergent scalar counterterm. It does not establish that every model containing additional fields must fail.

The coefficient field can also be included. Write $c(x) = 1 - S_{12}(x)$, where $S_n(2\cos(\theta)) = 2\cos(n\theta)$, and take $F(\theta, C) = V_{\text{lock}}(2\cos(\theta)) + \kappa[C - c(x)]^2$. On its tree valley, with kinetic scales f and g , the polynomial proportional to $\text{Tr}(M^4)/\Lambda^8$ is $[h + 2\kappa D(c')^2]^2/f^4 + 8\kappa^2 D(c')^2/(f^2 g^2) + 4\kappa^2/g^4$. The derivative is again reduced modulo P . All sixteen declared angular-only and two-scalar parameter cases give nonzero remainders. The calculation includes the coefficient scalar instead of silently integrating it out before evaluating the loops.

Finite scalar-loop displacements

$$\Delta V = \frac{m^4}{64\pi^2} \left[\log \frac{m^2}{\mu^2} - \frac{3}{2} \right], \quad \Delta\theta = -\frac{\Delta U'}{U''}$$

The scalar-only numerical calculation uses the local positive-mass $\overline{\text{MS}}$ expression $\Delta V = m^4 [\log(m^2/\mu^2) - 3/2]/(64\pi^2)$. The renormalized additional finite Wilson coefficients

are initialized to zero at the stated μ . Every example declares Λ/f , the lock coefficient and μ/m_{tree} . Resetting that boundary condition at another μ defines another example; it is not a claim that a fixed physical theory depends on the arbitrary renormalization scale after consistent running.

At $\theta_0=66$ degrees, $U''=12.2092424208$ and $U'''=-140.847974475$. The first-order angular displacement is $-\Delta U'(\theta_0)/U''(\theta_0)$. For $\Lambda/f=0.1$, no added lock, and $\mu=m_{\text{tree}}$, the numerical stationary point is approximately 65.997446321 degrees. The angle-based coefficient changes by about -0.2663 percent. The small-ratio displacement scales as $(\Lambda/f)^4$. The derivative and curvature are independently checked against finite differences, and the nonlinear stationary solution is compared with its linear prediction.

The sweep retains sixty angular examples. One leaves the physical positive-depth CKM chart; it remains in the results with a null CKM forecast rather than being dropped. Eighty of eighty-one declared two-scalar examples produce a controlled local stationary solution with a positive numerical Hessian. The unsuccessful case is retained with its parameters and failure reason. These calculations certify neither a global quantum minimum nor a cosmological occupation probability.

At the tree portal $C=1-2 \cos(12 \theta)$, $dC/d\theta=22.8253563911$ at the selected angle. A one-percent coefficient tolerance corresponds locally to about 0.00958804 degrees. Adding a large classical locking term is not automatically a radiative solution: it changes the mass and its derivatives, which also changes its own loop torque. The stored sweeps illustrate this effect directly.

A cubic supersymmetric constraint circuit

$$z_n = z^n, \quad \Phi_{60}(z) = z^{16} + z^{14} - z^{10} - z^8 - z^6 + z^2 + 1 = 0$$

Introduce sixteen gauge-singlet chiral fields $Z_1, \dots, Z_{16}$, with $Z_1=Z$, and sixteen drivers $D_2, \dots, D_{16}, D_{\text{star}}$. All Z fields have R charge zero; all drivers have R charge two. The declared CP action is complex conjugation and the chosen coefficients are real. A common real coupling can multiply the superpotential. With a real positive reference mass M , the base superpotential is $W = \sum_{n=2}^{16} D_n (M Z_n - Z_{(n-1)}) + M D_{\text{star}} [Z_{16} + Z_{14} - Z_{10} - Z_8 - Z_6 + Z_2 + M]$. Every term has mass dimension three.

At $D=0$, F -flatness gives $z_n = z^n$, where $z_n = Z_n/M$, and $\Phi_{60}(z) = z^{16} + z^{14} - z^{10} - z^8 - z^6 + z^2 + 1 = 0$. There are sixteen simple complex roots. They have unit modulus and are the primitive order-60 roots. This is an explicit renormalizable circuit implementing the algebraic equation; it avoids introducing a degree-sixteen superpotential interaction in the fundamental variables.

The constraint Jacobian J has nonzero determinant at every root. Its determinant divided by $\Phi_{60}'(z)$ is minus one in the declared ordering. Thus the chosen vacua are isolated. The full holomorphic mass matrix has blocks $[[0, J^T], [J, 0]]$, and every singular value of J occurs twice. With canonical kinetic terms, each pair has four real scalar modes and four fermionic spin modes with equal masses. The unbroken-SUSY one-loop contribution cancels at the F -flat zeros.

Perturbative protection here is the Wilsonian superpotential non-renormalization result for an initialized, globally supersymmetric sector, with a nonsingular positive Kahler metric. It does not force the initial coefficient choices. Kinetic corrections do not remove a zero of all F terms,

but they change spectra and matter when supersymmetry is broken. This distinction follows the Wilsonian treatment of non-renormalization, rather than treating a generic 1PI potential as an unchanged superpotential [2].

What the declared symmetry actually permits

For the base singlet sector, the complete holomorphic superpotential basis through field degree three has 2,448 monomials. R charge forces one driver, followed by zero, one or two neutral fields: $16(1+16+136)$. The selected circuit uses 37 monomials. CP and the declared R symmetry allow many other coefficients. They protect neither the chosen sparse pattern nor its relative integer values by themselves. The general CP-even quadratic Kahler metric contains 272 real parameters in the two neutral/driver blocks; the numerical mass calculations initialize it canonically.

A real change of the D_{star} tadpole changes $\Phi_{60}(z)$ to $\Phi_{60}(z)+\epsilon$. This deformation is allowed by the declared symmetries and changes both phase and radius. It demonstrates freedom in the initialized theory. It is not a claim that perturbative supersymmetric loops generate that absent term in the Wilsonian superpotential.

For a separate one-complex-field comparison, all CP-even operators $\text{Re}(z^a \bar{z}^b)$, $1 \leq a+b \leq 12$, are enumerated. There are 48 operators when no rotation symmetry is imposed; 42 have a nonzero carrier torque. The six desired harmonic numbers have gcd one. Among all cyclic rotations tested, orders two through sixty cannot permit every desired harmonic. This is a precise obstruction for that one-field CP-plus-cyclic class, not for arbitrary multi-field finite groups.

A holomorphic golden-coefficient extension

$$T = 2 - z^4 - z^6 + z^{14}, \quad T^2 - 3T + 1 = 0$$

The inverse phase needed for the original coefficient can be represented in the carrier without complex conjugation. Since $z^{60}=1$ on $\Phi_{60}=0$, $z^{-12}=z^{48}$. Exact reduction gives $T=1-z^{12}-z^{48}=2-z^4-z^6+z^{14}$ modulo Φ_{60} . It obeys $T^2-3T+1=0$. At the root with phase 66 degrees, $T=(3-\sqrt{5})/2$, with zero imaginary part.

Two additional chiral fields T and D_{gold} implement this linearly: $W_{\text{gold}}=M D_{\text{gold}}[T-2M+Z_4+Z_6-Z_{14}]$. T has R charge zero and D_{gold} has charge two. The extended model has 34 chiral fields and remains renormalizable. Its general allowed singlet superpotential through degree three contains 2,907 monomials. The extension fixes a golden spurion value in the selected initialized vacuum, while preserving the sixteenfold supersymmetric degeneracy. It does not fix a quark Wilson coefficient or explain why the twelfth power is the desired coefficient operator.

Actual field relaxation under energy selection

To lift the degeneracy, the worked scalar deformation is $V_{\text{soft}}/M^4=g_W^2 \epsilon \sum_n \text{Re}(z_n)$, with the six nonconstant Fourier coefficients a_n . This is an explicitly chosen supersymmetry-breaking scalar tadpole pattern. At exact circuit roots its leading energy is the original angular potential times $g_W^2 \epsilon$. The full fields are then allowed to relax; they are not constrained to remain equal to powers of a unit-modulus number.

For fixed z_1 , the other fifteen base neutral fields enter the F-term potential quadratically. They are eliminated by linear least squares, including the scalar tadpoles. The remaining complex z_1 is optimized in both its real and imaginary coordinates. All sixteen branches are continued for $\epsilon=10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}$. The branches descended from root powers eleven and forty-nine have the lowest energies in every declared continuation. Every retained branch has a positive reduced tree Hessian. This is a comparison of local continued vacua, not a proof that the full polynomial field potential has no distant competing vacuum or runaway.

At $\epsilon=10^{-4}$, the positive branch has phase 66.0005874177 degrees and radius 1.00002214970. The angle-only readout gives $C=0.382200029974$. The nonholomorphic real auxiliary readout $1-2 \operatorname{Re}(z_{12})$ gives 0.381955432748. The holomorphic extension gives $T=0.381669772701-0.000178406720 i$. These are different operators once the fields relax. A quark interaction must specify which operator it actually uses. Choosing a real auxiliary readout in a holomorphic superpotential would require a different construction; it cannot be substituted silently.

Full-circuit leading quantum calculation

$$A = J^\dagger J, \quad B = \sum_k \overline{F_k} H_k$$

The full quantum calculation includes the 34 chiral fields of the golden extension. On the R-symmetric $D=0$ slice, let F be the vector of driver constraints, J their Jacobian with respect to the seventeen neutral fields, and H_k their constant quadratic Hessians. Define $A=J^\dagger J$ and $B=\sum \text{conjugate}(F_k) H_k$. In canonical $\sqrt{2}$ real and imaginary neutral coordinates, the scalar squared-mass matrix is $[[\operatorname{Re}(A+B), -\operatorname{Im}(A)-\operatorname{Im}(B)], [\operatorname{Im}(A)-\operatorname{Im}(B), \operatorname{Re}(A-B)]]$. The driver scalar block has the corresponding unsplit singular-value spectrum.

The driver scalar trace cancels half of the fermion trace. The complete singlet one-loop potential therefore equals $g_W^4/(64 \pi^2)$ times $[\sum (34 \text{ neutral real modes}) b_i^2 (\log(g_W^2 b_i/\mu^2)-3/2)-2 \sum (17 \text{ Jacobian modes}) a_i^2 (\log(g_W^2 a_i/\mu^2)-3/2)]$. This expression accounts for all 68 real scalars and 34 Weyl fields, not just the light angular mode. It is evaluated in a DRbar example with canonical renormalized Kahler metric and specified scalar tadpoles at the reference scale. There are no gauge fields in this singlet-sector calculation.

The gradient uses eigenvector contractions with the analytic derivatives of A and B . It is independently checked against potential differences. At exact F-flat roots, the potential and gradient cancel to numerical roundoff. At the softly selected vacuum, the gradients need not cancel. The quantum stationary equations are solved in all thirty-four real neutral coordinates; the corrected neutral-sector Hessian is numerically positive. An independent expression for the complete 68-scalar, 34-Weyl effective potential also computes the quantum driver-block Hessian by finite differences. The unbroken common driver R phase makes the neutral-driver block vanish at zero drivers. Both corrected blocks have positive minimum eigenvalues in all nine declared cases. Step-size comparison checks the driver Hessian, while the agreement of the independent full and reduced potentials checks the mode counting. These are numerical local stability checks at one loop, rather than exact certificates or a higher-loop theorem.

Nine epsilon/coupling examples are retained: $\epsilon=10^{-6}, 10^{-4}, 10^{-3}$ and $g_W=0.1, 0.3, 1$. All produce controlled numerical neutral-sector minima. Leading tree-plus-one-loop energy

comparisons among all sixteen continued branches retain root powers eleven and forty-nine. At $\epsilon=10^{-4}$ and $g_W=0.3$, the phase becomes 66.0005860473 degrees, a quantum displacement of approximately -1.37037×10^{-6} degrees relative to the relaxed tree minimum. The coefficient field becomes $T=0.381670086337-0.000178010587 i$. The largest actual neutral-coordinate quantum displacement is about 4.67×10^{-7} . These are chosen weak-coupling examples, not experimental uncertainties or a derived cosmological scale.

Quark operators: hierarchy without an enforced coefficient

The declared holomorphic quark sector uses two flavor charges. The left-handed doublets Q_i have charges (1,1),(0,1),(0,0), while u_i^c and d_i^c have the opposite charges. H_u, H_d and all neutral circuit fields have zero flavor charge. A has charge (-1,0), B has charge (0,-1), and Q, f^c have R charge one. The scalar A and B backgrounds remain free inputs. No anomaly-free gauged flavor completion or mechanism fixing their magnitudes is claimed.

All allowed operators $Q_i H f_j^c A^a B^b$ times neutral monomials are enumerated through superpotential field degree six. There are 6,460 operators with the sixteen base neutral fields, and 7,560 when T is included. The up and down sectors have independent real coefficients before spontaneous CP breaking. The pure singlet enumeration and this Yukawa enumeration are complete within their declared sectors and degree bounds; they are not an enumeration of all Standard Model, lepton or higher-Kahler interactions.

In particular, $Q_1 H d_2^c A$, $Q_2 H d_3^c B$, and $Q_1 H d_3^c A B$ carry independent coefficients. Allowing their spurion powers does not relate those coefficients. A T insertion can multiply the third operator, but its independent coefficient and the lower-degree A B operator remain allowed. Neutral fields can also appear in diagonal and up-sector entries. The phase of the scalar vacuum therefore does not automatically become the standard CKM phase. Higher-Kahler interactions such as $Q_1^\dagger Q_2 A^\dagger/M$ are allowed and can alter canonical normalization.

A derived texture counterexample

$$V_{\text{CKM}} = U_u^\dagger U_d, \quad U_f^\dagger Y_f Y_f^\dagger U_f = \text{diag}(y_f^2)$$

The worked trial uses $Y_u = \text{diag}(u_1, u_2, u_3)$ and an upper-triangular Y_d with entries $Y_{12}=a d_2$, $Y_{23}=b d_3$ and $Y_{13}=c a b \exp(-i \theta) d_3$. The diagonal spectra are declared positive numerical inputs. This is a selected member of the allowed operator family. Up-sector off-diagonal terms and most neutral insertions are initialized to zero; the symmetry does not forbid them.

The matrices are diagonalized through the eigenvectors of $Y Y^\dagger$. For each independent c , only a and b are matched to the two existing anchors $|V_{us}|=0.22431$ and $|V_{cb}|=0.0411$. The values of $|V_{ub}|$ and CP observables are then consequences of the matrix, without fitting them. At $c=\phi^{-2}$ and $\theta=66$ degrees, the texture gives $|V_{ub}|=0.00360299463$ and effective depth 0.390811666690, rather than the nominated chart's 0.00352144218 and 0.381966011250. Its standard CKM phase is approximately 65.9786293 degrees. Finite diagonalization effects matter even in this simple hierarchical matrix.

Changing the independent coefficient while retaining the same two anchors changes $|V_{ub}|$ substantially. The stored six-coefficient sweep supplies explicit examples. This is a mathematical obstruction to inferring the exact relation $s_{13}=C s_{12} s_{23}$ from charge

arithmetic alone. A stronger quark construction must relate Wilson coefficients, kinetic normalization and matrix diagonalization, or predict the correction instead of absorbing it into a fit.

Scale evolution as a separate matching test

$$16\pi^2 \frac{dY_u}{d\log\mu} = \left[\frac{3}{2}(H_u - H_d) + \alpha_u I \right] Y_u$$

The implementation also integrates one-loop Standard Model matrix Yukawa equations using $t=\log(\mu)$, left-row Yukawa convention and SU(5)-normalized g_1 . In this convention $\beta_u = [(3/2)(H_u - H_d) + T_{\text{gauge},u} I] Y_u / (16\pi^2)$, with the corresponding down and lepton equations and the one-loop gauge equations. The source's $\log(\mu^2)$ convention is converted explicitly [3]. Rephasing invariance, a common-left-basis covariance check and the top-only limit independently check the implementation.

The example initializes the original exact CKM chart with declared illustrative spectra and gauge couplings. At $\mu/\mu_0=10^8$, its effective depth differs from ϕ^{-2} by about -8.62 times 10^{-5} relatively, and its standard phase differs by about -0.00349036 degrees. The absolute $|V_{ub}|$ also changes as the anchors run. No measured matching scale is supplied. This is a standalone SM evolution diagnostic, not a threshold-matched evolution of the supersymmetric circuit. It shows that a claim of an exact flavor relation must identify the scale at which it is imposed.

What is now supported and what remains physical

The exact supports are the carrier algebra, the stationary-family identities, the global-winner preservation argument, the scalar-counterterm remainder and Bezout witness, and the complete declared finite operator spaces. The supersymmetric protection is a conditional application of the established Wilsonian non-renormalization result to the supplied globally supersymmetric circuit. The mass spectra, relaxed fields, one-loop extrema, quark diagonalizations and SM evolution are numerical computations with explicit parameters and limitations. No new Lean theorem is claimed.

The constructed model answers whether an initialized protected equation can be implemented by renormalizable fields and whether a specified weak selector produces a controlled nearby vacuum. It does so. The outstanding physics is the origin of the sparse initialized coefficients and selector terms, a quark interaction enforcing or replacing the nominated exact mixing rule, consistent kinetic and threshold matching, higher-loop stability, and a vacuum history. The branch comparison does not choose the positive CP sign. A complete theory of strong CP or baryogenesis has not been supplied.

Reproduction and retained calculations

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_flavor_completion.py` from the repository root, then the corresponding `plot_flavor_completion.py`. Run `unittest discovery with test_flavor_completion.py` for the new mathematical and numerical checks, or `test_flavor*.py` for the combined flavor checks. NumPy and SciPy support numerical spectra, extrema and running; Matplotlib supports the scientific figure. Exact polynomial identities and operator labels use rational arithmetic and the existing PerfectPower algebra.

The receipts retain all declared cases, including one uncontrolled two-scalar continuation and one angular example outside the positive-depth CKM chart. The two exhaustive quark-operator tables are stored as deterministic gzip-compressed JSON; ordinary gzip decompression recovers the complete tables. The manifest covers the scientific JSON outputs, including those compressed tables. Fresh-copy reproduction checks those outputs in the same numerical runtime. A byte-identical replay is evidence of reproducibility in that environment; the tests provide separate mathematical and numerical checks. PDF generation uses ReportLab and standard installed fonts. Source code, prior repository work, raw receipts, diagrams and the monograph are included in the complete handoff archive.

The preceding vacuum-completion run completed 841 tests successfully, with four skips. Its 25 new completion tests cover the exact carrier obstruction, circuit elimination and mode pairing, independent loop gradients, corrected driver Hessian, complete declared operator enumeration, texture counterexamples and scale-evolution covariance. A fresh archive extraction reproduced all twenty scientific receipt, manifest and figure files byte for byte in the declared runtime. No new Lean theorem was added in that push.

Primary references

- [1] S. Coleman and E. Weinberg, Radiative Corrections as the Origin of Spontaneous Symmetry Breaking, *Physical Review D* 7, 1888 (1973).
<https://doi.org/10.1103/PhysRevD.7.1888>. This supplies the effective-potential framework; the worked scalar and circuit schemes are stated explicitly above.
- [2] N. Seiberg, Naturalness Versus Supersymmetric Non-renormalization Theorems (1993),
<https://arxiv.org/abs/hep-ph/9309335>. The relevant distinction is protection of a chosen Wilsonian superpotential, with separate kinetic and supersymmetry-breaking effects. It does not derive the chosen flavor coefficients.
- [3] A. V. Bednyakov, A. F. Pikelner and V. N. Velizhanin, Three-loop SM beta-functions for matrix Yukawa couplings (2014), <https://arxiv.org/abs/1406.7171>. The one-loop matrix conventions and scale conversion are the parts used here.
- [4] I. de Medeiros Varzielas, D. Emmanuel-Costa and P. Leser, Geometrical CP violation from non-renormalisable scalar potentials (2012), <https://arxiv.org/abs/1204.3633>. This is a comparison for symmetry-constrained higher-order operators preserving particular vacua; it does not establish the present order-60 construction.

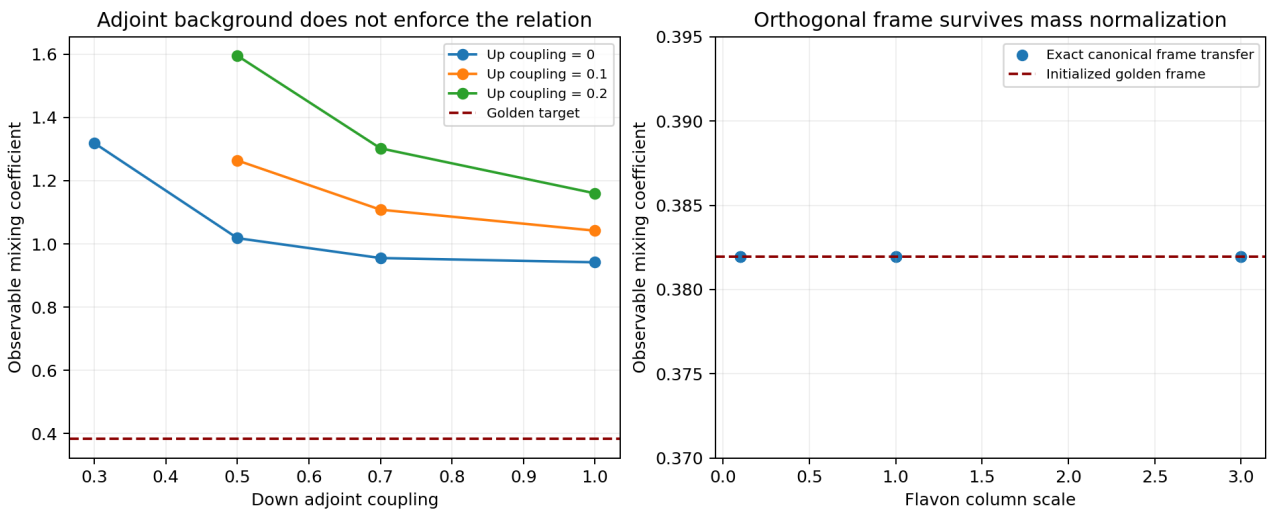
Shared mediators and the alignment boundary

What this mediator push establishes

The next step now has an explicit non-Abelian quark mediator construction and an exact tree matching calculation that includes the light right-handed kinetic normalization. The family symmetry makes each heavy mediator mass proportional to the identity and gives one contraction for each flavon column. Separately orthogonal flavon frames then determine the CKM matrix independently of the six column magnitudes and the two flavor-universal mediator masses. This is a constructive quark interaction that transmits an initialized mixing frame without the finite-mass diagonalization errors of the previous triangular texture.

It does not independently derive that frame. The accompanying scalar analysis gives a precise obstruction to the simplest alignment route. For six fundamental flavons with separate phase symmetries, fixed positive norms and exactly orthogonal up and down frames, every CP-even orientation term through scalar degree six reduces to a constant plus a linear function of $|V_{ij}|^2$. An exact rational assignment certificate proves that a permutation always attains its global minimum on the frame manifold. A generic nondegenerate cost therefore cannot select the required hierarchical CP-violating mixing. Degenerate costs can leave mixed flat directions; those are not a unique prediction.

I also tested a more favorable alternative in which an adjoint mediator mass contains initialized neighboring mixings, a golden 1-3 entry and the 66-degree phase. Both up and down adjoint couplings are allowed independently by the symmetry. Exact tree matching and canonical normalization were performed before extracting CKM quantities. Thirty of thirty-six declared cases satisfy the numerical control criteria. Every accepted case fails the golden target by more than one percent, despite matching the two existing anchors. The six unsuccessful or uncontrolled cases are retained with their reasons. The construction is therefore a tested candidate and a boundary result, not a completed predictive theory of flavor.



The exact observable target

$$C_{\text{eff}} = \frac{w(1-w^2)}{uv} = \phi^{-2}$$

Write $u=|V_{us}|$, $v=|V_{cb}|$ and $w=|V_{ub}|$. The nominated physical relation is $C_{\text{eff}}=w(1-w^2)/(uv)=\phi^{-2}$, where $\phi=(1+\sqrt{5})/2$. This follows from the original standard-angle rule $s_{13}=C s_{12} s_{23}$ because $u=s_{12} c_{13}$, $v=s_{23} c_{13}$ and $w=s_{13}$. It is already rephasing invariant. Relating a Yukawa matrix entry to other entries is insufficient unless its diagonalization yields this observable equation.

For $p=u^2$, $q=v^2$ and $r=w^2$, define $A=r(1-r)^2$. Eliminating C from $C^2-3C+1=0$ gives $A^2-7pqA+(pq)^2=0$. The positive physical branch $C_{\text{eff}}<1$ selects ϕ^{-2} ; the other positive root is ϕ^2 . The equation does not fix a CP phase. In raw fixed-norm flavon overlaps its formal maximum scalar degree is twenty-four. Squaring it into an energy term would encode the desired result by construction and would not explain its origin.

The existing anchors remain $u=0.22431$ and $v=0.0411$. The small-branch target is $w=0.00352144218276$. No new experimental data or fit to $|V_{ub}|$ or CP measurements was introduced. All numerical examples use declared illustrative quark spectra and dimensionless mediator units. A claim of an exact relation still needs a specified matching scale and subsequent threshold and running calculations.

An exact Abelian obstruction for the initialized circuit

Consider an additional ordinary diagonal Abelian phase action preserving every monomial of the initialized singlet superpotential, with its mass M and numerical coefficients neutral. The recursion terms force $q(Z_n)=n q(Z_1)$ and $q(D_n)=-n q(Z_1)$. The constant D_{star} term forces $q(D_{\text{star}})=0$. Its remaining terms require $n q(Z_1)=0$ for $n=2,6,8,10,14,16$. Their greatest common divisor is two, so the general character relation is $2 q(Z_1)=0$.

The golden extension has a constant D_{gold} term, forcing $q(D_{\text{gold}})=0$. Its linear $D_{\text{gold}} T$ term then forces $q(T)=0$. Hence an operator O and $O T$ have the same additional charge. Such a charge rule cannot forbid $Q_1 H d_3^c A B$ while allowing $Q_1 H d_3^c A B T$. The implementation derives the monomial constraints from the existing circuit rather than a separate assumed polynomial, and checks cyclic groups of orders one through 120. The focused test independently extends those checks through order 240.

This result is deliberately restricted. Charged mass or coupling spurions, symmetries mixing fields, and non-Abelian representations can change the premises. Treating the initialized coefficients as transforming spurions would supply a different model and would require its own alignment and coupling audit. The obstruction does not rule out flavor symmetry in general.

The declared non-Abelian mediator sector

Use the Standard Model gauge group, a formal global $SU(3)_F$ family symmetry, six independent $U(1)$ flavon phase symmetries and CP. Q is a family anti-triplet. The three right-handed-conjugate up fields and the three right-handed-conjugate down fields are family singlets. There are six complex gauge-singlet fundamental flavons F_{ui} and F_{di} . Each carries its own phase charge $+1$, while its associated right-handed-conjugate quark carries -1 . The Higgs and Q have zero phase charge.

Introduce heavy X_u and X_d family anti-triplets and X_u^c and X_d^c family triplets. X_u and X_u^c are vectorlike color-triplet weak-singlet fermions with hypercharges $+2/3$ and $-2/3$. X_d and X_d^c have hypercharges $-1/3$ and $+1/3$. Each family triplet gives three heavy Dirac species, for six species overall. These pairs are vectorlike under the Standard Model. The global family group is not claimed to be an anomaly-free gauged completion; its Goldstone and cosmological questions are outside this calculation.

In ordinary left-handed Weyl notation, the allowed core interactions are $M_f X_f \text{ dot } X_f^c$, $h_f Q \text{ dot } H_f X_f^c$ and $\sum_i \lambda_{fi} X_f \text{ dot } F_{fi} f_i^c$, plus their Hermitian conjugates. H_u is the Higgs doublet and H_d is its conjugate doublet. This notation is an ordinary non-SUSY mediator model. It is not a combined softly broken completion of the previously constructed 34-chiral-field singlet sector.

The operator audit exhausts fermion mass bilinears and two-fermion-one-scalar Yukawa invariants in this declared quark sector. It checks Standard Model gauge representations, family representations and all six phase charges. With the seventeen neutral scalar backgrounds Z_1 through Z_{16} and T , including their conjugates, there are 78 independent invariant coefficients. Two are bare universal masses, two are Higgs vertices, six are flavon-column vertices and 68 are neutral-scalar universal-mass vertices. CP permits real coefficients; it does not equate them. Lepton interactions and a complete scalar potential for all neutral backgrounds are outside this finite audit.

Adding one complex phase-neutral family adjoint Σ and its conjugate adds four invariants, giving 82. They are $X_u \Sigma X_u^c$, $X_u \Sigma^\dagger X_u^c$, $X_d \Sigma X_d^c$ and $X_d \Sigma^\dagger X_d^c$. Up and down coefficients are independent. In the Hermitian background used below the two coefficients in each sector combine into one real number, but the symmetry does not impose any relation between the sectors or between an adjoint coupling and a singlet mass.

Canonical tree matching without a small-flavon expansion

$$A = M^{-1}L, \quad K = I + A^\dagger A, \quad Y = -hAK^{-1/2}$$

Let L_f be the 3 by 3 matrix of flavon columns multiplied by their λ_{fi} couplings, and let M_f be the invertible heavy mediator mass matrix. At zero Higgs background the heavy fermion row block is $[L_f, M_f]$. Define $A_f = M_f^{-1} L_f$, $K_f = I + A_f^\dagger A_f$ and $R_f = K_f^{-1/2}$. A normalized basis for the three light right-handed fields is $N_f = [R_f; -A_f R_f]$. Direct multiplication gives $[L_f, M_f]N_f = 0$ and $N_f^\dagger N_f = I$.

The Higgs vertex projected into that normalized null space gives $Y_f = -h_f A_f R_f$. This is exact at tree order at unbroken electroweak matching, to all orders in the ratio of flavon mixing to heavy mass. It is not the unnormalized expression $-h_f M_f^{-1} L_f$. No loop threshold is included. At finite Higgs vacuum value, higher-dimensional Higgs operators and mixing with weak-singlet heavy fermions also affect the charged current. The dimension-four effective CKM matrix is unitary; the finite-Higgs full theory need not give an exactly unitary light 3 by 3 submatrix.

The focused checks verify the normalized null frame and Yukawa projection directly. They also compare against independent singular values of the complete 6 by 6 fermion mass matrix $[[0, h_f v_H I], [L_f, M_f]]$. The three light singular values divided by v_H converge to those of the matched Y_f . A separate finite-Higgs test compares the light charged-current overlaps

with the matched CKM limit and checks their small departure from unitarity. Noncommuting mass matrices are tested under a common family-basis rotation, so the checks do not rely only on diagonal mediator masses.

What universal masses can and cannot transmit

For a universal mass $M_f = m_f I$ and a general singular decomposition $L_f = U_f \text{diag}(s_{fi}) W_f^\dagger$, the canonical Yukawa matrix has the same left eigenvectors U_f . Its singular values are $|h_f| s_{fi} / \sqrt{|m_f|^2 + s_{fi}^2}$. This map is strictly increasing for positive s_{fi} , preserving nondegenerate mass ordering. Thus changing a flavor-universal mediator mass, even by a complex golden or cyclotomic background, cannot change the mixing angles or physical CKM phase of fixed L_u and L_d . This statement does not require orthogonal flavon columns.

With separately orthogonal columns, $L_f = U_f \text{diag}(\ell_{fi})$, the stronger separation holds: changing each individual column magnitude only changes its associated quark singular value. While retaining the assigned nondegenerate mass ordering, the CKM matrix is $U_u^\dagger U_d$ and is independent of those magnitudes and the universal masses. Twelve declared examples initialize $U_u = I$ and U_d from the original golden CKM chart, then vary column magnitudes and real or complex universal masses. They retain the nominated coefficient and 66-degree standard phase to numerical precision. These are conditional transfer checks. The down frame was copied from the target chart and has not been derived from a symmetry-controlled potential.

A separate three-case test uses nonorthogonal random columns and changes only universal masses. It confirms the general unchanged-eigenvector result while the spectra change. Both results narrow the physical role a golden field can play: placing it only in a universal heavy mass does not orient the CKM matrix. Placing its phase as an overall Yukawa-column multiplier also fails, because $(Y \text{diag}(\exp(i \theta_{aj}))) (Y \text{diag}(\exp(i \theta_{aj})))^\dagger = Y Y^\dagger$. A physical CP phase must influence relative left-handed directions or interfering flavor structures.

The complete low-degree triplet alignment basis

For the six complex triplets alone, write $G_{ab} = F_a^\dagger F_b$. Independent phase invariance requires each triplet label to have equal occurrences with and without conjugation. At scalar degree two there are six norms. At degree four there are 21 products of norms and fifteen squared overlaps $|G_{ab}|^2$, giving 36 quartic invariants. Including the quadratic terms, the renormalizable triplet potential has 42 independent real coefficients. One Standard Model Higgs adds its quadratic term, its quartic term and six Higgs-norm/flavon-norm portals, giving fifty terms in that restricted scalar sector.

At degree six the CP-even basis has 56 triple products of norms, ninety norm-times-squared-overlap terms and twenty real Gram triangles $\text{Re}(G_{ab} G_{bc} G_{ca})$ with distinct labels. Products of balanced $SU(3)$ epsilon tensors are Gram determinants and reduce to this basis. Unbalanced epsilon contractions are forbidden by the six separate phase symmetries. The first Gram-rank determinant relation for six triplets in three dimensions has degree eight, so it does not remove any of these degree-six contraction types.

Now restrict to positive fixed norms and exactly orthogonal U_1, U_2, U_3 and D_1, D_2, D_3 frames. Overlaps within each frame vanish. Every three-label triangle contains two labels from

one frame, so every Gram triangle vanishes. Norm products are constant. The surviving squared cross overlaps are proportional to $B_{ij}=|U_i^\dagger D_j|^2=|V_{ij}|^2$. Norm factors at degree six only modify their coefficients. Therefore the complete orientation dependence through degree six is $E=\text{constant}+\sum_{ij} K_{ij} B_{ij}$.

Fixed norms and exact separate orthogonality are assumptions of this theorem. Arbitrary finite quartic coefficients in the full potential can distort the frames, and a model outside this regime requires a new calculation. An adjoint introduces new orientation tensors and is also outside this theorem. Neutral singlets can change norm-dependent parameters; multiplying a quartic overlap by a singlet would exceed renormalizable degree and does not create a new degree-four orientation tensor.

A rational global certificate for the orientation obstruction

$$E = \sum_i \alpha_i + \sum_j \beta_j + \sum_{ij} R_{ij} B_{ij} \geq E_{\text{permutation}}$$

For any unitary relative frame, B is doubly stochastic: its entries are nonnegative and each row and column sums to one. For each declared rational cost matrix K , the implementation finds row and column numbers α_i and β_j with $\alpha_i + \beta_j \leq K_{ij}$. It also finds a permutation attaining the same sum $\sum_i \alpha_i + \sum_j \beta_j$. Define reduced costs $R_{ij} = K_{ij} - \alpha_i - \beta_j$, which are exactly nonnegative rational numbers.

Then $E = \sum_i \alpha_i + \sum_j \beta_j + \sum_{ij} R_{ij} B_{ij}$ is bounded below by the permutation energy. A permutation matrix is itself unitary, so the bound is attained within the physical frame manifold. The receipt supplies the rational dual numbers, reduced costs and all winning permutations. Tests check eighty further rational cost matrices independently against all six assignments and compare the identity against nontrivial random unitaries.

This proves that a permutation is always a global minimizer of the restricted linear orientation energy. With a unique minimum assignment, no nonpermutation B attains the same energy. For degenerate costs, a mixed CP-violating frame can instead belong to a flat minimum set. The constant-cost example explicitly retains such a direction. The correct conclusion is that this potential cannot uniquely select the nominated nonpermutation CKM matrix in the stated regime, not that every possible minimum must have zero CP violation.

The first higher-degree CP capability example

Scalar degree eight permits $|F_{ui}^\dagger F_{dj}|^4$ and real four-overlap cycles. With equal fixed norms, the chosen positive sum of fourth powers reduces to $\sum_{ij} B_{ij}^2$. Each row obeys $\sum_j B_{ij}^2 \geq 1/3$, so the total is at least one. Equality requires every $B_{ij} = 1/3$. The unitary three-dimensional Fourier frame reaches the bound and has $|J| = 1/(6 \sqrt{3})$, with a conjugate CP partner.

This is an explicit CP-even orientation interaction that can select nonzero physical $|J|$ when paired with nondegenerate quark column magnitudes. It selects democratic mixing, with every CKM magnitude $1/\sqrt{3}$, and fails the hierarchical target. Its coefficient pattern is chosen; other degree-eight invariants are independently allowed under the declared symmetry. No renormalizable UV generation of that pattern or connection to the protected 66-degree singlet vacuum has been provided. The example demonstrates what new orientation dependence becomes possible at degree eight, while leaving the actual golden alignment unresolved.

The favorable adjoint candidate and its failure

The numerical candidate uses a Hermitian traceless adjoint background Σ with entries $\Sigma_{12}=a$, $\Sigma_{23}=b$ and $\Sigma_{13}=\phi^{-2} a b \exp(-i \theta)$, together with their conjugates and zero diagonal. θ is initialized at 66 degrees. The mediator masses are $M_u=I+g_u \Sigma$ and $M_d=I+g_d \Sigma$. The columns before filtering are diagonal, using the same illustrative hierarchies as the preceding quark calculation. This gives the golden factor and phase a direct place in a flavor-dependent mediator, which a universal mass could not supply.

The assumed adjoint background is deliberately favorable and explicitly initialized. No scalar potential producing it is claimed. Both g_u and g_d are independent allowed couplings. Canonical tree matching is performed for each sector, followed by exact numerical diagonalization of $YY^\dagger$. Only a and b are matched to $|V_{us}|$ and $|V_{cb}|$. They are not tuned to $|V_{ub}|$ or any CP observable.

The declared sweep uses $g_u=0,0.1,0.2$; $g_d=0.3,0.5,0.7,1$; and column scales 0.1,1,3. A case is accepted only when the two anchor residuals are below 10^{-9} , both mediator matrices have minimum singular value at least 0.2 and $\max(a,b) \leq 1$. Thirty cases are accepted and six fail the matching or control criteria. Accepted C_{eff} ranges from approximately 0.941676 to 1.595694, against the golden target 0.381966. All accepted cases fail the target at the one-percent level. The input phase also does not become the standard CKM phase automatically. Four conjugate-pair checks retain magnitudes and reverse J .

Unnormalized Yukawa predictions are retained for the same backgrounds so that the kinetic correction can be inspected. They are not substituted for the canonical result. The failure is not a theorem against all adjoint models; it demonstrates that this favorable entry pattern and the allowed shared-mediator couplings do not enforce the desired observable relation. A successful adjoint construction would need a derived vacuum and additional relations among its allowed couplings, followed by the same normalization and mixing checks.

Reproduction, validation and the next interaction target

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_flavor_mediator.py`, followed by `python python/plot_flavor_mediator.py` with the same environment. The runner uses only the existing $|V_{us}|$ and $|V_{cb}|$ means. It retains exact charge, operator and rational dual certificates, the conditional transfer cases, all adjoint attempts and the scientific figure. NumPy and SciPy support the numerical matching; the exact charge and assignment statements use integer or rational arithmetic. There are no new Lean theorems or newly acquired experimental datasets.

The twenty focused tests cover the charge obstruction, independent operator-list construction, exact rational assignment witnesses, canonical null-space matching, the complete fermion mass comparison, family-basis covariance, CP conjugation, universal-mass invariance and the failed golden enforcement. They accompany the repository-wide suite and a fresh-extraction byte comparison of the retained scientific artifacts. The integrated repository-wide run completed 879 tests successfully, with four skips. A fresh archive extraction reproduced all 32 scientific receipt, manifest and figure files from both flavor investigations byte for byte. The full test and replay results are recorded in the validation receipt for this push.

The resulting research target is more specific than adding a golden Yukawa coefficient. The quark mediator interaction can already transmit an orthogonal frame exactly, so the remaining positive construction is a symmetry-controlled relative frame that produces the golden amplitude relation and the physical CP phase. Within the tested phase-symmetric triplet class, a uniquely selected nonpermutation frame requires new orientation dependence beyond degree six or a change of fields or assumptions. Higher-degree interactions, additional non-Abelian representations or nonorthogonal-frame dynamics are possibilities to test; none is asserted here to solve the problem. Threshold matching, running, a consistent combined singlet/flavor theory and a vacuum history remain separate requirements.

Primary comparisons and their limits

S. Antusch, S. F. King, M. Malinsky and M. Spinrath, Quark mixing sum rules and the right unitarity triangle, <https://arxiv.org/abs/0910.5127>. Its texture-zero sum-rule route motivates deriving mixing from both Yukawa sectors. It does not establish the present golden relation.

R. Alonso, M. B. Gavela, L. Merlo and S. Rigolin, On The Potential of Minimal Flavour Violation, <https://arxiv.org/abs/1103.2915>. Its analysis provides context for restrictions of low-degree dynamical flavor potentials. The six-triplet fixed-frame obstruction above has separate explicit premises and its own rational certificate; no claim of duplicating every model in that paper is made.

J. Gehrlein, J. P. Oppermann, D. Schafer and M. Spinrath, An $SU(5) \times A_5$ Golden Ratio Flavour Model, <https://arxiv.org/abs/1410.2057>. It provides an example of an explicit renormalizable flavor and messenger construction. Its golden mixing result concerns the neutrino sector, not our nominated CKM coefficient. Only the primary abstracts and metadata were used for these comparisons.